

Master Thesis Project

Update Meeting, 07.05.2026

Starting Point

The Art of Making Problems Simple: A Theory of Intelligence

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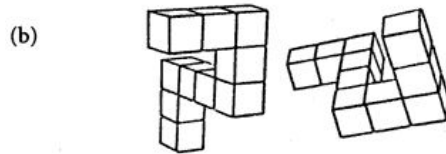
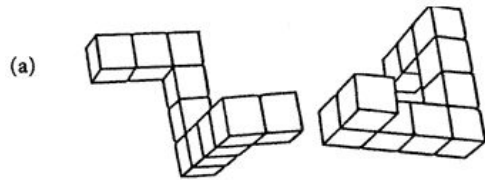
Abstract

Human intelligence is flexible and general, yet its underlying structure remains poorly understood. We propose that intelligence arises from the ability to construct and revise representational normal forms, that is, coordinate systems in which prediction, comparison, and inference become simple. This capacity rests on four operations: identifying lawful generators of variation, quotienting task-irrelevant differences, constructing measurable axes for ordered reasoning, and selecting representations that locally flatten the space of possibilities.

“intelligence is ability to construct and revise representational normal forms”

- (1) discovering lawful generators of variations in the world
- (2) quotienting task-irrelevant distinctions
- (3) ...

Application



Mental Rotation Test—Are these two figures the same except for their orientation?

	$g_0 = e$	$g_1 = R_\theta \circ S_a$	$g_2 = g_1 \circ g_1$
$o_1 = \text{trumpet}$			
$o_2 = \text{monkey}$			
$o_3 = \text{drill}$			?



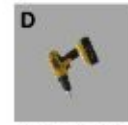
(o_3, g_2)



(o_1, g_2)



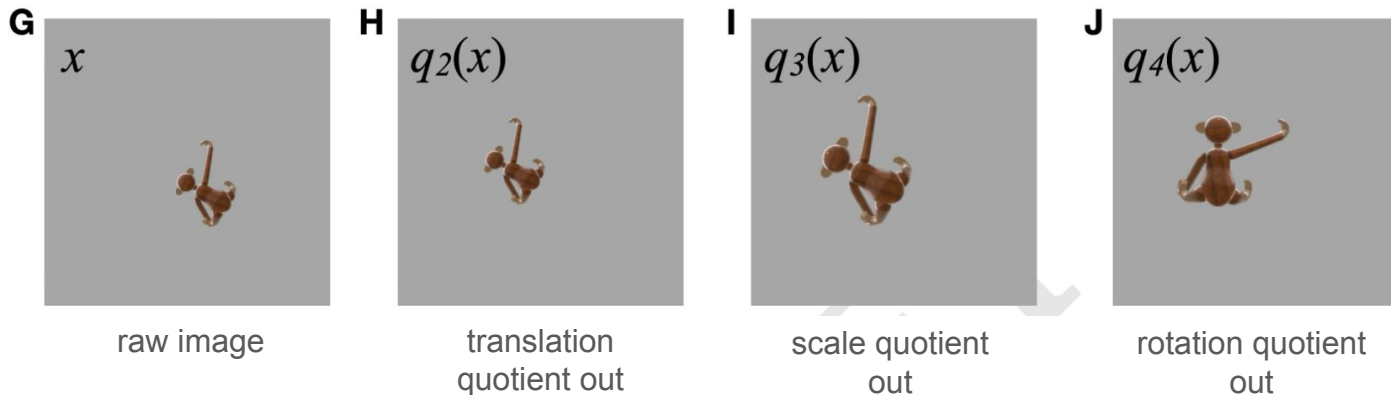
(o_3, g_{other})



$(o_3, g_1 \circ R_\theta)$

Project's Goal

already built:



Project's goal: incorporate the other rotation axis

Dataset

3 shapes



Banana



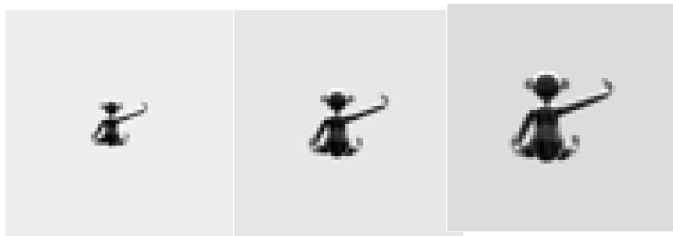
Kettle



Toy Monkey

- 64 x 64 pxs each
- grayscale

3 scales



3 x 3 translations



Dataset

10 roll rotations (x-axis)



10 pitch rotations (y-axis)



10 yaw rotations (z-axis)



3 shapes x 3 scales x (3x3) translation x 10^3 orientations = **81.000 images**

Stage 4 - Route A (Prototype orbit approach)

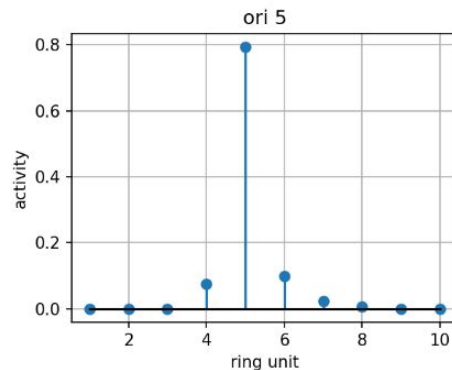
1. Encode canonicalized object into a set of **orientation prototypes** $\{\mu_1, \mu_2, \dots, \mu_K\}$



2. Build **shape-specific decoder**

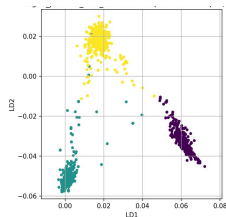


$$p_t = \text{softmax}(Wq_t + b)$$



Stage 4 - Route A (Prototype orbit approach)

- Learn **orbit dynamics** with shift operator $(Gp)_i = k_{-1}p_{i-1} + k_0p_i + k_{+1}p_{i+1}$
with step consistency $p_{t+1} \approx Gp_t \quad p_t \approx G^{-1}p_{t+1} \quad p_{t+k} \approx G^k p_t$
- Train **two networks** with a classifier **to construct quotient space**
with object identity



Latent representation as: $z = (z_{\text{trans}}, z_{\text{scale}}, p_{\text{rot}}, q_{\text{quot}})$

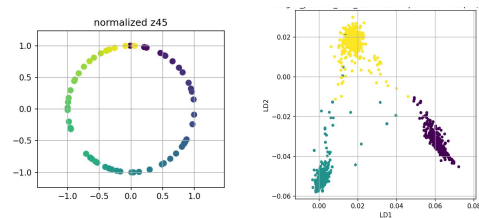
Stage 4 - Route B (Continuous vector-feature approach)

1. Build orbit prototypes
2. Build **transitions graph** between prototypes using spectral embedding

$$\theta_i = \text{atan2}(v_3(i), v_2(i))$$

3. Train a **network head to discover this circular embedding** and at the same time the quotient space with a classifier network

$$q_t \mapsto (u_t, v_t, q_{\text{quot}})$$

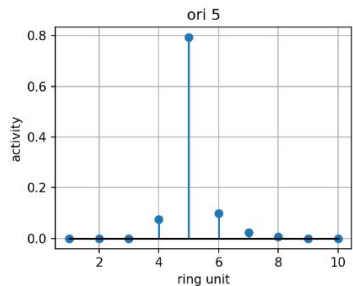


Latent representation as: $z = (z_{\text{trans}}, z_{\text{scale}}, z_{\text{rot}}, q_{\text{quot}})$

Stage 4 - Comparison

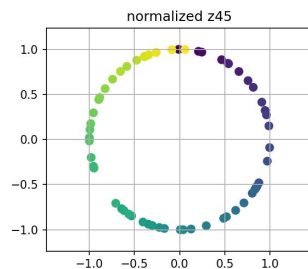
Route A (Prototype orbit approach)

- orientation as a **bump distribution on a discrete ring**
- output: $p_t \in \mathbb{R}^K$
- orientation as **activity over ring units**
- decoder: similarity to orientation prototypes



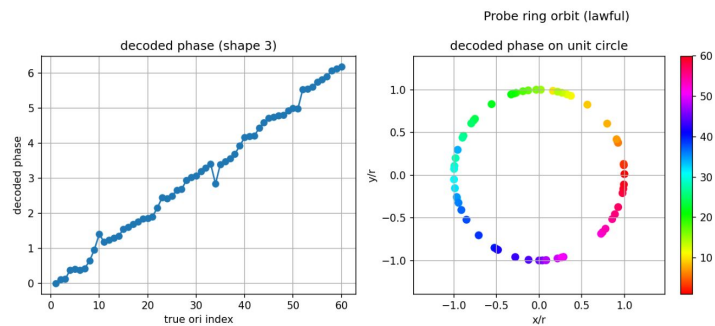
Route B (Continuous vector-feature approach)

- orientation as a **continuous 2D circular coordinate**
- output: $(u_t, v_t) \in \mathbb{R}^2$
- orientation as **position on a learned circle**
- decoder: learned neural mapping

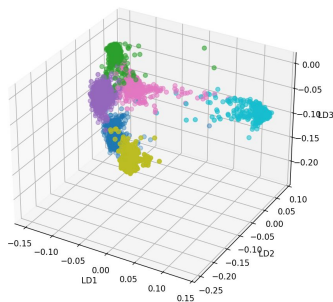


Stage 4 - Comparison

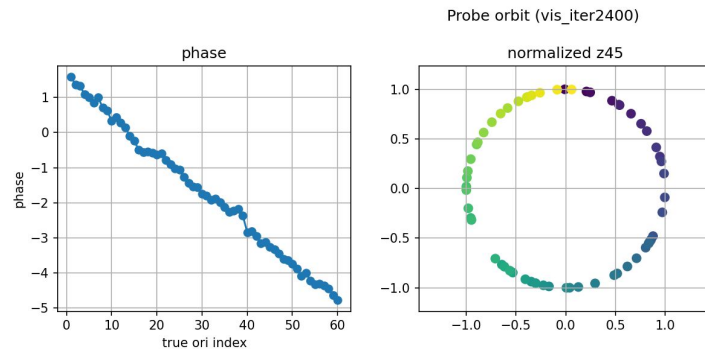
Route A (Prototype orbit approach)



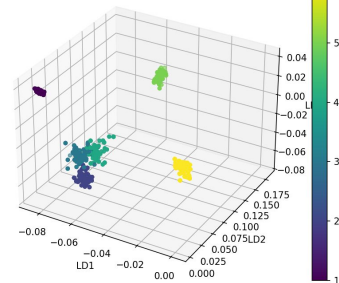
q4 cluster separation (LDA 3D), acc=0.993



Route B (Continuous vector-feature approach)



stage4_posthoc_CAN: LDA3(q) color=shape (CAN)



Stage 4 - 3D

Route A (Prototype orbit approach)

Build **shape-specific decoder** for **every rotation axes**

$$p_t = \text{softmax}(Wq_t + b)$$

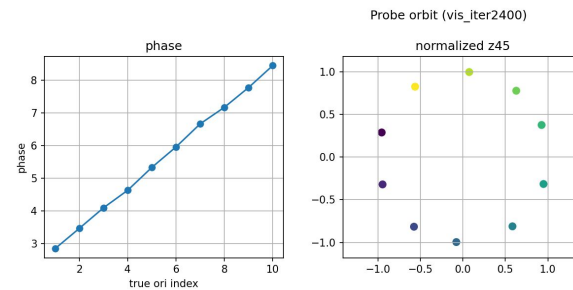
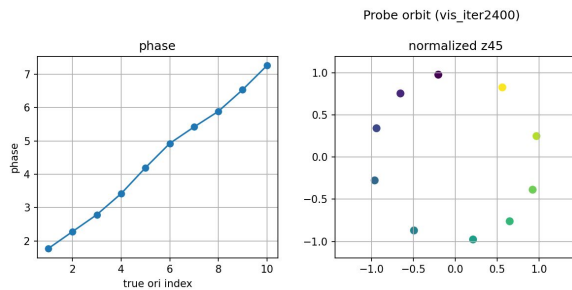
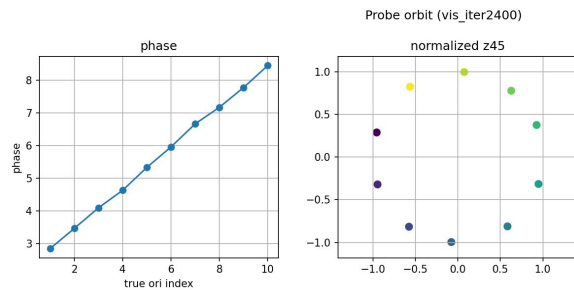
Route B (Continuous vector-feature approach)

Sequentially train 3 networks, one for every rotation

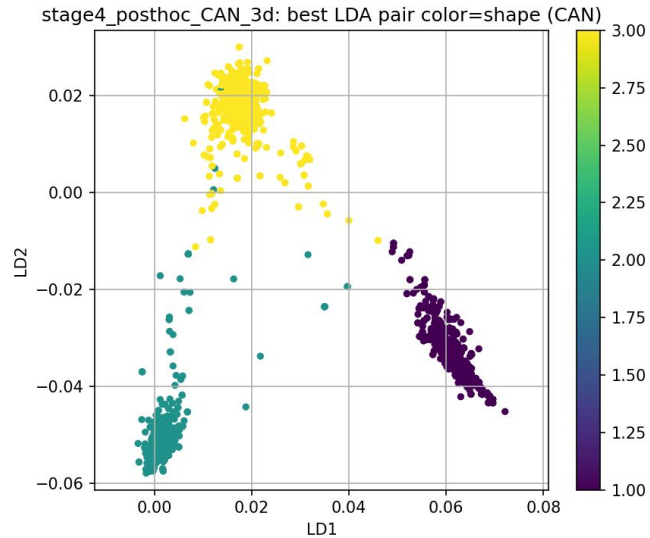
$$q_t \mapsto (u_t, v_t, q_{\text{quot}})$$

$$z = (z_{\text{trans}}, z_{\text{scale}}, p_{\text{roll}}, p_{\text{pitch}}, p_{\text{yaw}}, q_{\text{quot}}) \quad z = (z_{\text{trans}}, z_{\text{scale}}, z_{\text{roll}}, z_{\text{pitch}}, z_{\text{yaw}}, q_{\text{quot}})$$

Route B (Continuous vector-feature approach)

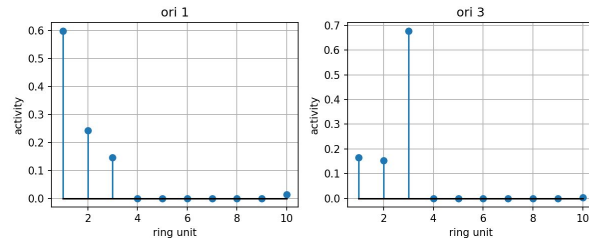
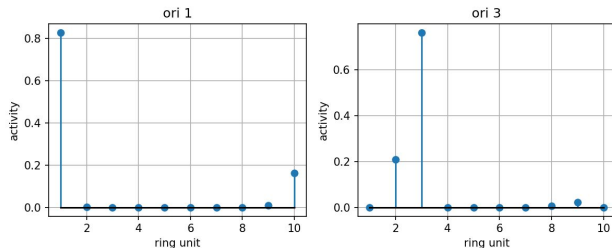
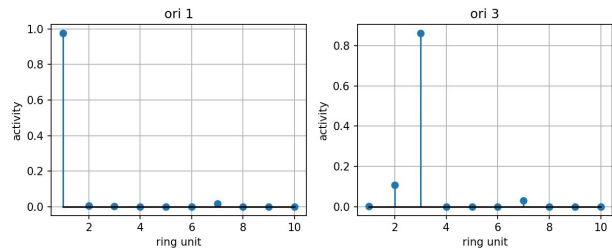
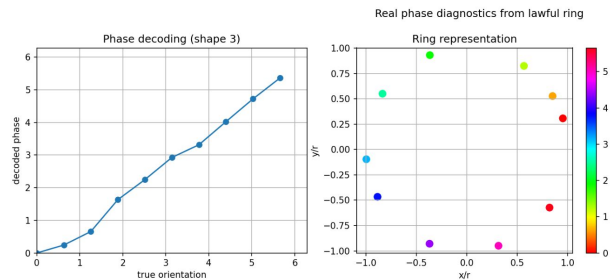
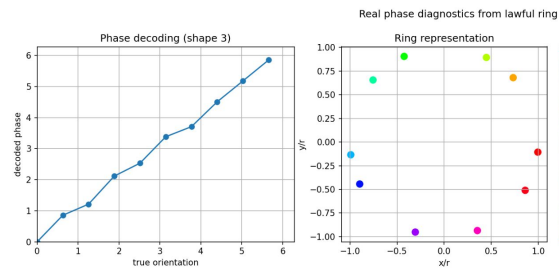
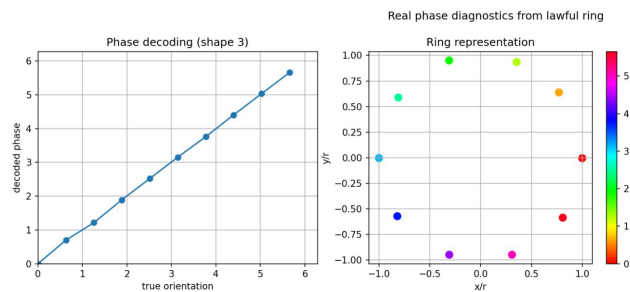


Route B (Continuous vector-feature approach)

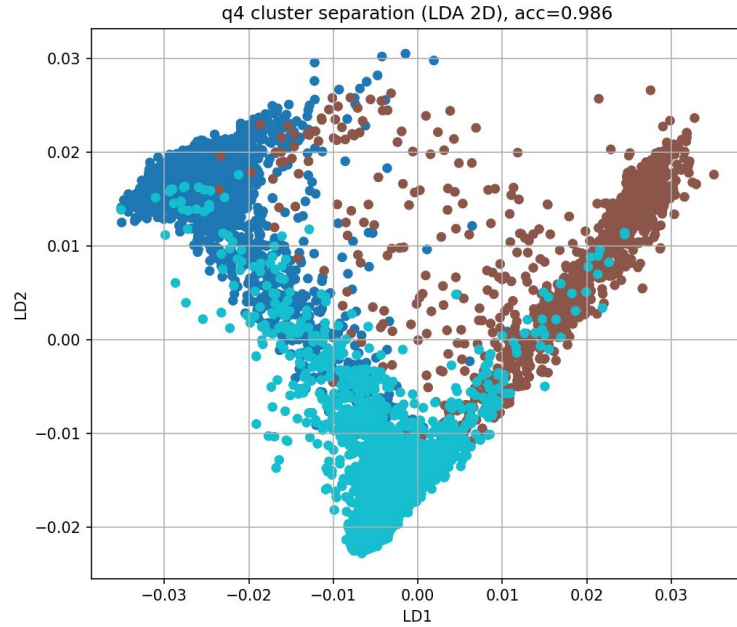


separability score at 0.985

Route A (Prototype orbit approach)



Route A (Prototype orbit approach)



separability score at 0.986